

Unfolding Complex Knowledge Graphs in Large Mixed Reality Space

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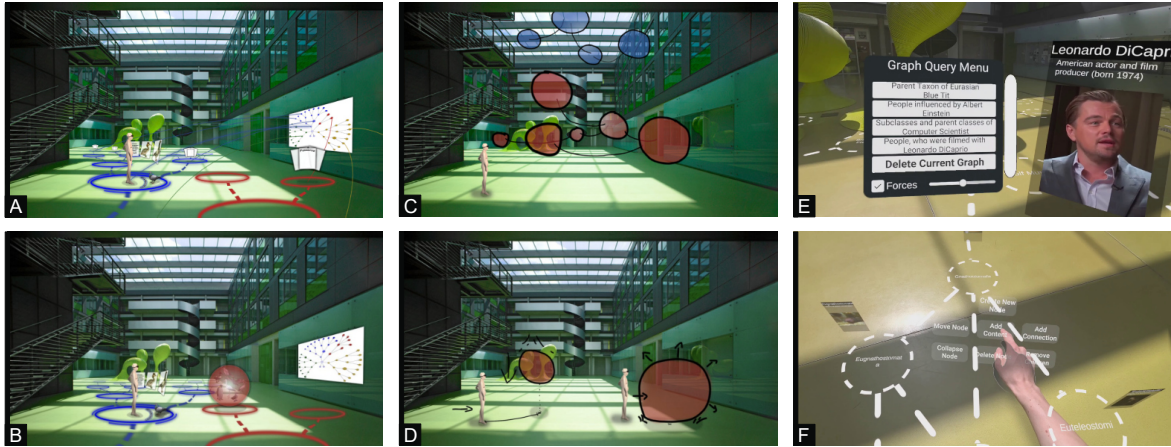
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Figure 1: Overview of the proposed method for exploring large scale knowledge graph data in mixed reality with floor-based layout ((A) and (B)) and mid-air layout ((C) and (D)). As well as the method for immersive in situ graph data authoring ((E) and (F)).

ABSTRACT

With the advent of mixed reality technologies, low-cost mobile head-mounted displays such as the Meta Quest series have become widely affordable. These devices come with markerless inside-out tracking capabilities and onboard operating systems and thus can be used in large spaces for interactive data exploration. In this work, we introduce design considerations, concepts, and methods to enhance the interactive exploration and authoring of knowledge graph datasets within architectural-scale mixed reality environments. We have developed a prototype showcasing the potential of our concepts. Moving forward, we plan to continue developing the system and conduct user studies to further evaluate our concepts.

Index Terms: Knowledge Graph Data Exploration, Mixed Reality

1 INTRODUCTION & BACKGROUND

Knowledge graphs, which represent information through interconnected nodes and edges, are powerful tools for visualizing complex relationships and data structures. Such graphs have been used in various fields to support knowledge representation, wikidata is one of the well-known examples [5]. However, traditional 2D interfaces for exploring these graphs might fail to convey the depth and complexity of the information they contain.

One of the most compelling benefits of Mixed Reality (MR) is its ability to create highly immersive environments that significantly increase *user engagement*. Traditional methods of presenting information might struggle to maintain attention and interest, particularly in environments that require passive consumption of

data. Besides, in MR environments, users can explore knowledge graphs in three dimensions, allowing them to move around the data, view it from different perspectives, and interact with it in a spatial context. This helps users to better perceive the structure, hierarchy, and connections between nodes. Such immersive exploration enables a *more comprehensive understanding* of complex relationships within the graph data. Furthermore, unlike traditional flat-screen navigation, MR users can potentially “touch” or “walk into” the dataset directly for zooming or manipulating, resulting in a more intuitive and engaging *interaction experience*. In addition, MR can incorporate *multimedia elements* such as 3D models, animation, and audio to create a multi-sensory data exploring experience. These elements not only make the content more engaging but also cater to different exploration styles, ensuring that a wider audience can benefit from the material. For example, visual learners can benefit from detailed 3D models, while auditory learners can follow along with explanatory audio.

While some previous work has demonstrated the feasibility of rendering graph datasets directly in AR or VR environments, certain challenges must be overcome to better support immersive data exploration. Firstly, existing work tends to render a complete knowledge graph data in space without offering methods for natural navigation [1, 2], the experience remains hindered when the data scales. Guidelines are needed to support effective display and exploration of data. Secondly, beyond exploration, the immersive environment also serves as an ultimate auxiliary for further organizing the data [6]. The design of immersive interfaces that offer knowledge graph data authoring experience is also desired.

In this work, we aim to support a more intuitive and engaged experience for immersive graph data exploration by designing novel immersive interfaces. Our main contributions are the following: (1) Data layout concepts for engaged interactive data viewing, and in situ authoring concepts that support intuitive modification of knowledge graph data. (2) A prototypical system interfacing with the wikidata dataset to demonstrate the concepts.

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2 DESIGN CONSIDERATIONS AND CONCEPTS

2.1 Design Considerations

Avoid Overwhelm and Encourage Exploration. When exploring large-scale knowledge graph datasets, displaying all the data at once is impractical due to rendering limitations and cognitive overload for users. Instead, a potentially more effective approach is to start from a specific node and explore the graph incrementally by discovering and unfolding nearby nodes, following paths of interest.

Intuitive Navigation. Besides displaying knowledge graphs statically in virtual environments, the data navigation system could leverage the user's physical movement for a more interactive viewing experience. For example, designs could allow users to fold and unfold data nodes in knowledge graphs by simply walking around.

Body Anchored Interaction for Authoring. To create a more engaging node placement authoring process, we propose empowering users to directly manipulate nodes and links through immersive UI canvases and their physical movements, akin to the lens authoring process proposed in [3].

2.2 Layout Concepts

2.2.1 Floor-embedded Layout

The "Circles of Knowledge" layout design leverages the metaphor of connected circles on the floor, each representing a distinct node within the knowledge base (Fig. 1 (A)). Users stand within these circles to reveal dynamic, floating information panels with rich content, including images and text, providing detailed information for the specific node.

Upon entering the mixed reality environment, users find themselves surrounded by interconnected circles on the floor. Each circle signifies a knowledge node, such as a key concept, topic, or category. By stepping into a circle, users activate a floating information panel that hovers at eye level. These panels dynamically present multimedia content related to the selected node.

To enhance the representation of unfolded categories, this design incorporates the rendering of transparent spheres to enclose 3D models or images of items within the category (Fig. 1 (B)). When a user selects a category, the corresponding sphere unfolds to reveal the contents.

2.2.2 Mid-Air Layout

In this concept, users are immersed in an environment where knowledge nodes are manifested as interconnected spheres. Each sphere encapsulates rich multimedia content.

Upon entering the space, users encounter an array of floating spheres (Fig. 1 (C)), each representing a distinct knowledge node. These spheres are connected by visible threads, forming a three-dimensional network. For interaction, we adopted a balloon metaphor: users physically move through the space, when standing below a sphere, the sphere then descend like balloons being pulled down by a string to reveal detailed information related to the specific node (Fig. 1 (D)).

2.3 Authoring Concepts

While exploring the graph data, we applied a force-directed algorithm to automatically distribute the nodes across the space. However, for more advanced exploration, there is a need to allow users to organize the nodes according to their preferences.

In this work, we implemented functions along with immersive 3D UIs to help users organize nodes and links with CRUD operations directly in situ. The system allows users to add, delete, update (the content, position, direction), as well as expand & collapse nodes and links (as shown in Fig. 1 (F)).

3 INTERACTIVE PROTOTYPE

We implemented our proof-of-concept system using ViewR [4], a Unity plugin that provides essential functions for delivering large-scale multi-user mixed reality experiences. As some example queries shown in Fig. 1 (E), we interfaced our MR system to the wikidata dataset [5] with SPARQL querying language. Within the system, all visualizations and interactions are implemented and optimized for real-time usage. Moving forward, voice and gesture-based interactions could be involved and specifically designed.

4 CONCLUSION & FUTURE DIRECTIONS

In this work, we propose concepts and methods for immersive in situ exploration of knowledge graph data in mixed reality. A proof-of-concept prototype is implemented and usability is confirmed with a test run. The source code of our prototype will be available on the project page.

In the future, apart from continuing to develop the system, the layout concepts could be further extended by adapting to the physical shapes and characteristics of surrounding objects, such as different types of furniture. Additionally, user studies could be conducted to evaluate the system's effectiveness across diverse user groups and various spatial structures or scales.

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